

Curriculum Vitae – Brandon Zhao

byzhao@caltech.edu

EDUCATION

Caltech

September 2021 - Present

Ph.D., Computing and Mathematical Sciences

Pasadena, CA

- **GPA:** 4.2/4.0
- **Advisor:** Katherine L. Bouman
- **Relevant Coursework:** Machine Learning and Data Mining, Linear Analysis with Applications, Mathematical Optimization, Probability Theory and Stochastic Processes

Duke University

August 2017 - May 2021

B.S., Mathematics and Computer Science (Double Major)

Durham, NC

Minor: Statistical Science

- **GPA:** 3.9 / 4.0, Dean's List (x4)
- **Relevant Coursework:** Graduate Machine Learning, Design/Analysis of Algorithms, Applied Stochastic Processes, Advanced Probability, Measure and Integration, Statistical Inference

RESEARCH EXPERIENCE

PhD Student Researcher

June 2026 – Present

Google Deepmind – Neural Radiance Fields Group

San Francisco, CA

- Working on large-scale 3D generative models

Research Assistant

September 2021 - Present

Caltech Computing and Mathematical Sciences

Pasadena, CA

- Currently conducting research in computational imaging, with a focus on astrophysical applications

PhD Student Researcher

June 2025 – April 2025

Google Deepmind – Neural Radiance Fields Group

San Francisco, CA

- Developed novel neural architectures for use in physics-informed neural networks (PINNs)

Undergraduate Researcher

May 2019 – June 2021

Duke Computer Science Department – CODE+ Program

Durham, NC

- Designed interpretable prototype-based neural network to classify images from large datasets into fine-grained subcategories
- Investigated different ways to interpret and improve network decisions with visualizations and addition of different prototype regularizations
- Increased performance and interpretability of previous model through addition of recurrent, explainable attention mechanism and secondary fine-grained prototype network

SURF Research Fellow

June 2020 - August 2020

NASA JPL – Machine Learning and Instrument Autonomy Group

Pasadena, CA (Remote)

- Designed and trained neural networks for finding and classifying objects of interest in unlabeled Mars Rover panoramic camera images using transfer learning technique
- Created custom multi-label dataset both for algorithm training and for use as a guideline for future human-labeling efforts
- Explored interpretable techniques for improving minority class performance with supervised learning through supplementary image annotations

Research Assistant

May 2018 - July 2018

Duke Mathematics Department – DOMath Program

Durham, NC

- Investigated properties of epidemic models on random graphs, specifically periodic trees, in small group environment

SELECTED PUBLICATIONS

“Video reconstruction of variable VLBI observations with neural fields.”

Marianna Foschi*, **Brandon Zhao***, Antonio Fuentes*, Katherine L. Bouman, José Luis Gómez, Aviad Levis

Conditionally Accepted, *Nature*

“Fourier Feature Pyramids for Physics-Informed Neural Networks.”

Brandon Zhao, Yixuan Wang, Jonathan T. Barron, Katherine L. Bouman, Dor Verbin, Pratul P. Srinivasan

In Submission

“Generative Diffusion Priors for 3D Mapping of the Dark Universe.”

Brandon Zhao, Diana Scognamiglio, Olivier Doré, Katherine L. Bouman

CVPR 2026

"Mapping Dark-Matter Clusters via Physics-Guided Diffusion Models."

Diego, **Brandon Zhao**, Adolfo Muñoz, Diego Gutierrez, Katherine L. Bouman.

ECCV 2026

“Revealing the 3D Cosmic Web with Gravitationally Constrained Neural Fields”

Brandon Zhao, Aviad Levis, Liam Connor, Pratul P. Srinivasan, Katherine L. Bouman

ICLR 2025

“Single-View Refractive Index Tomography with Neural Fields”

Brandon Zhao, Aviad Levis, Liam Connor, Pratul P. Srinivasan, Katherine L. Bouman

CVPR 2024

“This Looks Like Those: Illuminating Prototypical Concepts Using Multiple Visualizations”

Chiyu Ma*, **Brandon Zhao***, Chaofan Chen, Cynthia Rudin

NeurIPS 2024

"Mars Image Content Classification: Three Years of NASA Deployment and Recent Advances"

Kiri Wagstaff, Steven Lu, Emily Dunkel, Kevin Grimes, **Brandon Zhao**, Jesse Cai, S.B. Cole, Gary Doran, Raymond Francis, Jake Lee, and Lukas Mandrake

The Thirty-Third Annual Conference on Innovative Applications of Artificial Intelligence (IAAI-21).